Show, Attend and Tell: Neural Caption Generation with Visual Attention

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INTRODUCTION

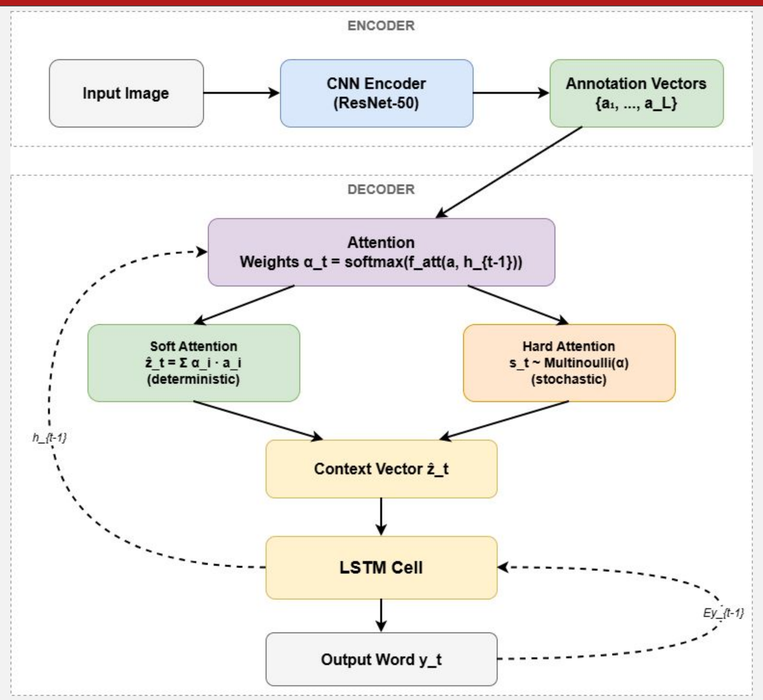
Problem: Image captioning requires CV (understanding images) and NLP (generating sentences).

Goal: Reproduce the "Show, Attend and Tell" (Xu et al., 2015).

Key Idea: Attention lets the model focus on different image regions while generating each word.

Example image and captions from Flickr8k Dataset:
"A beagle and a golden retriever wrestling in the grass."
"Two puppies play in the grass."

METHODOLOGY



Soft attention: Fully differentiable. Trained with backpropagation on cross-entropy loss.

Hard attention: Samples one location. Non-differentiable, so trained with REINFORCE w/ variance reduction

$$
\begin{aligned}
\nabla_W L_s \approx\ & \nabla_W \log p(y \mid \tilde{s}, a) && \text{(supervised)}\\
& + \lambda_r \left(\log p(y \mid \tilde{s}, a) - b\right) \nabla_W \log p(\tilde{s} \mid a) && \text{(REINFORCE)}\\
& + \lambda_e \nabla_W H[\tilde{s}] && \text{(entropy reg.)}
\end{aligned}
$$

where:

$\tilde{s} \sim \text{Multinoulli}(\alpha)$ — sampled attention trajectory

y — target caption; a — image annotation vectors

b — moving-average baseline of recent log-likelihoods

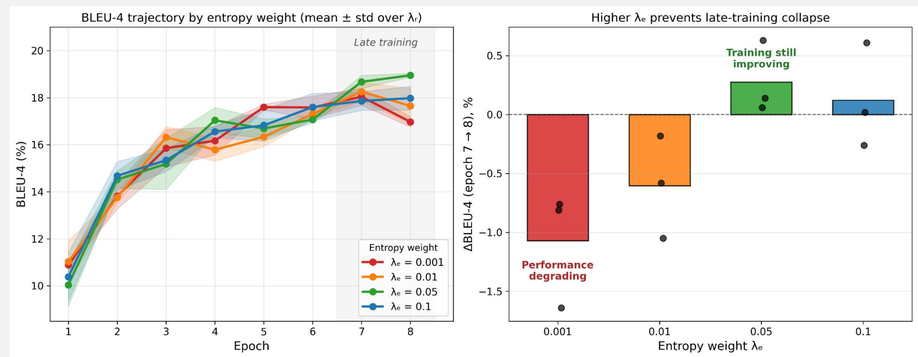
$H[\tilde{s}]$ — entropy of the attention distribution

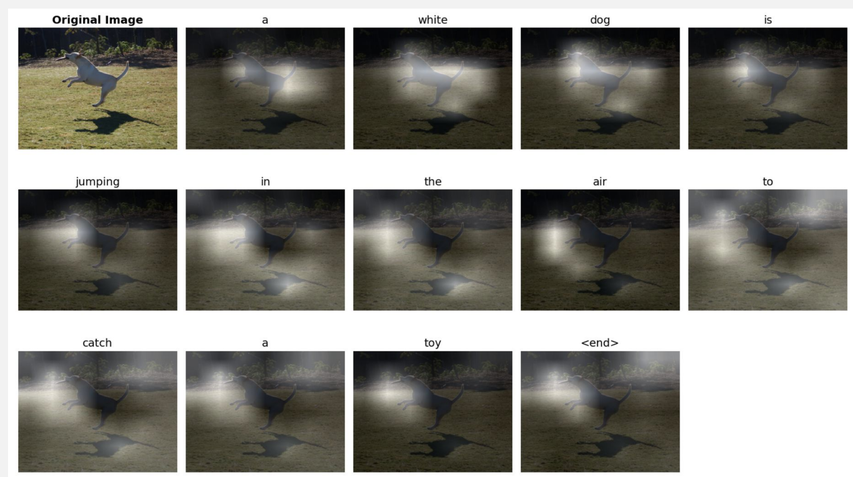
λ_r, λ_e — REINFORCE and entropy weighting hyperparameters

RESULTS

Model	BLEU-1	BLEU-2	BLEU-3	BLEU-4
Paper — Soft Attention	**67.0**	44.8	29.9	19.5
Paper — Hard Attention	**67.0**	45.7	31.4	21.3
Ours — Soft Attention	60.0	43.2	29.9	19.4
Ours — Hard Attention	60.4	43.3	30.1	20.7
Ours — ViT + LSTM	60.5	43.4	30.4	21.2
Ours — ViT + Transformer	61.9	**45.1**	**32.2**	**22.7**

Results on Flickr8k: ViT + Transformer is the best of our tested models, exceeding the paper on BLEU-3 and BLEU-4.



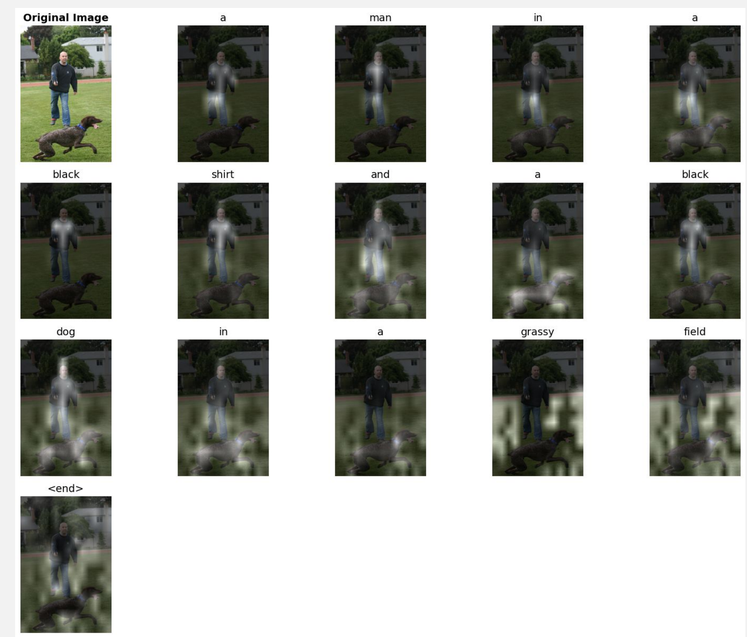


Attention maps during caption generation. Illustrates where the model attends at each decoding step.

In the failure case, attention maps reveal the source of the error — the model fixates on the red **NO SKIING** sign for both "red" and "sled". Shows the model is looking at the right thing but lacks the vocabulary to label it correctly, defaulting to the more common Flickr8k concept "sled." Attention separates *perception* errors from *language* errors.

EXTENSION

1. Vision Transformer Encoder (ViT)
   a. Replaced CNN encoder with ViT-B/16
   b. Used pretrained ImageNet weights and freezed encoder for stable training
2. Transformer Decoder
   a. Replaced LSTM with transformer decoder
   b. Self-attention for better long range context in sentence
   c. Multi-Head Cross Attention for attending to image regions when generating each word



Attention maps when training using ViT. More discrete due to the use of patches.

CONCLUSION

We successfully reimplemented the paper Show, Attend and Tell, reproducing its idea of using attention to align image regions with generated words. Building on this foundation, we introduced a ViT encoder and a Transformer decoder, upgrading the model while preserving the original attention mechanism.

Future work could focus on improving performance by fine tuning the ViT encoder instead of freezing it, allowing better adaptation to the captioning task. We can also train on larger datasets like MS COCO to improve generalization.

REFERENCES

K. Xu, J. Ba, R. Kiros, K. Cho, A. Courville, R. Salakhutdinov, R. Zemel, and Y. Bengio, "Show, attend and tell: Neural image caption generation with visual attention," in *Proc. 32nd Int. Conf. Mach. Learn. (ICML)*, vol. 37, Lille, France, 2015, pp. 2048–2057. [Online]. Available: https://proceedings.mlr.press/v37/xuc15.html